Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
AI

Practical - IV

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Class: TYIT	Batch: B1
Date of Assignment: 25/07/2026	Date/Time of Submission: 25/07/2026

Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.
Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

```
# Probabilities

P_Python = 0.20

P_Java = 0.80

P_Placed_Python = 0.90

P_Placed_Java = 0.40


# Total probability of being placed

P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)


# Bayes Rule
```

```
P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed

P_Java_Placed = (P_Placed_Java * P_Java) / P_Placed


print("Probability student knows Python:", P_Python)

print("Probability student knows Java:", P_Java)

print("Probability placed student knows Python:", round(P_Python_Placed, 4))

print("Probability placed student knows Java:", round(P_Java_Placed, 4))


print("Pranay T003")
```

Output:

```
Probability student knows Python: 0.2
Probability student knows Java: 0.8
Probability placed student knows Python: 0.36
Probability placed student knows Java: 0.64
Pranay T003
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:

```
# Probabilities

P_Blue = 0.15

P_Green = 0.85

P_Correct = 0.80

P_Wrong = 0.20
```

```
# Probability that the witness says Blue

P_Witness_Blue = (P_Correct * P_Blue) + (P_Wrong * P_Green)


# Bayes' Theorem

P_Blue_Given_WitnessBlue = (P_Correct * P_Blue) / P_Witness_Blue


print("Probability witness says Blue:", round(P_Witness_Blue, 4))

print("Probability taxi was actually Blue given the witness said Blue:",

    round(P_Blue_Given_WitnessBlue, 4))


print("Pranay T003")
```

Output:

```
Probability witness says Blue: 0.29
Probability taxi was actually Blue given the witness said Blue: 0.4138
Pranay T003
```